

SAKSHAM UPADHYAY

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Education

VIT Bhopal University

2022 - Present

Bachelor of Technology in Electronics and Communication Engineering - 8.4/10

Bhopal

De Paul School

2021 - 2022

Intermediate 82

Singrauli

De Paul School

2019 - 2020

Secondary 78

Singrauli

Projects

1. Price prediction models (Flight and House) with Explainable AI

- Developed a Price Prediction Model using XGBoost and CatBoost with 85% accuracy on a Kaggle dataset of 30,00,00+ records for flight and 500+ records for house.
- Created a **Streamlit-based frontend** enabling users to interact with **2+ prediction models**, improving accessibility and user experience for **100+ test cases**.
- Incorporated **Explainable AI (XAI)** techniques like SHAP to interpret model predictions, enhancing transparency and trust with **clear insights for 85%+ of predictions**.
- Integrated **3+ category-specific models** into a unified Streamlit interface, ensuring **scalability and easy expansion** to support **5+ additional use-cases**.

2. Chat with PDF

- Engineered a multilingual **AI PDF chatbot** using LangChain and Google Gemini API, capable of processing and answering questions from **100+ page documents** in **4+ languages** including English, Spanish, French, and German.
- Executed a conversational pipeline integrating **GoogleGenerativeAI embeddings** with **FAISS vector search**, enabling **semantic retrieval from 1000+ text chunks** for accurate responses.
- Built real-time **speech-to-text and text-to-speech** features using **SpeechRecognition** and **gTTS**, supporting **voice-based queries and audio responses** for accessibility.
- Deployed a **Streamlit-based UI** with persistent chat history, chat export, and user feedback system, resulting in a seamless experience across **10+ end-to-end user interactions**.

3. Deep Fake Audio Detection

- Architected a CRNN model achieving 73%+ accuracy** in detecting audio deepfakes with 3 convolutional layers and a 2-layer bidirectional LSTM, trained on 1,000+ real/fake audio samples.
- Optimized audio preprocessing** by standardizing variable-length inputs to fixed 200-frame spectrograms (16kHz SR, 1024 n_fft), reducing inference latency by 30% through GPU-accelerated batch processing (16 samples/batch).
- Reduced overfitting by 15%** via dropout (0.3) and adaptive learning rate scheduling (patience=2), while maintaining 90%+ validation accuracy across 10 training epochs.
- Implemented handwriting recognition to interpret and solve drawn numbers and mathematical expressions.
- Focused on clean, modular code for scalability, cross-platform compatibility, and user-friendly functionality.

Technical Skills

Languages: Java, Python, JavaScript, HTML

Developer Tools: VS Code

Technologies/Frameworks: React, Tailwind, Express Js

AI/ML and NLP Tools: Generative AI, Natural Language Processing (NLP), LangChain

Soft Skills: Problem-solving, Collaboration, Communication

Certifications

- HTML, CSS, and Javascript for Web Developers
- Java certification by SkillVertex